

2013 HKDSE Physics

Paper 2 Elective Sections

Original questions with official answers and English explanations. Correct answers are red; explanations are blue.

Section A Astronomy and Space Science

Question 1.1

Arrange these celestial bodies in ascending distance from Earth: (1) the Sun; (2) Sirius, 8.6 ly from Earth; (3) Uranus, 19 AU from Earth.

- A. (1) (2) (3)
- B. (1) (3) (2)
- C. (3) (1) (2)
- D. (3) (2) (1)

Official answer: B

B. The Sun is about 1 AU away. Since $1 \text{ ly} \approx 63\,200 \text{ AU}$, 8.6 ly is far greater than 19 AU: Sun, Uranus, Sirius.

Question 1.2

Which statements about Ptolemy's geocentric and Copernicus's heliocentric models are correct?

- (1) Orbits are circular in both models.
- (2) Earth is at the centre of the Moon's orbit in both models.
- (3) Both models can explain retrograde motion.

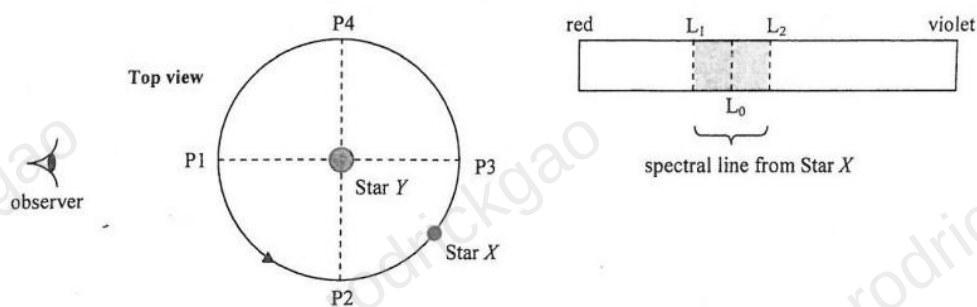
- A. (1) only
- B. (3) only
- C. (1) and (2) only
- D. (1), (2) and (3)

Official answer: D

D. Both historical models use circular motions and keep the Moon orbiting Earth. Ptolemy uses epicycles for retrograde motion; in the heliocentric model it follows from relative orbital motion.

Question 1.3

Star X orbits star Y approximately in a circle in the direction shown. An Earth observer finds a spectral line varying between L_1 and L_2 ; its laboratory wavelength is L_0 . Which wavelengths correspond to P1, P2, P3 and P4, respectively?



A. L_0, L_1, L_0, L_2

B. L_1, L_0, L_2, L_0

C. L_0, L_2, L_0, L_1

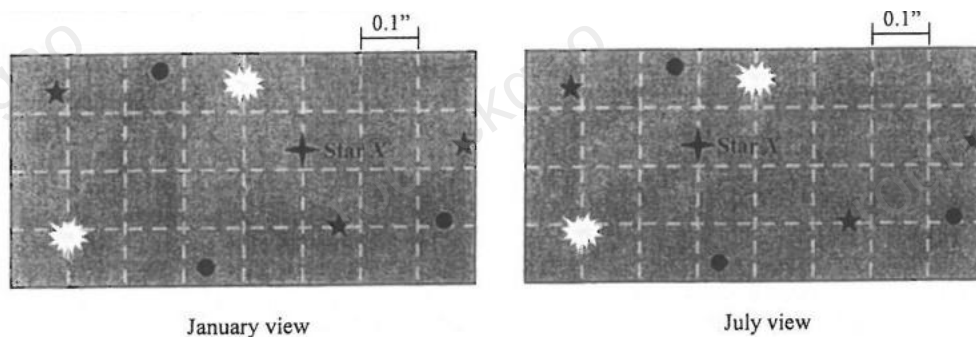
D. L_2, L_0, L_1, L_0

Official answer: A

A. At P1 and P3 the velocity is perpendicular to the line of sight, so there is no radial Doppler shift. At P2 X recedes, giving the redder L_1 ; at P4 it approaches, giving the bluer L_2 .

Question 1.4

These photographs of the same sky region were taken six months apart. Each grid square subtends 0.1 arc second. What is the distance of star X in parsecs?



A. 0.1 pc

B. 0.2 pc

C. 5 pc

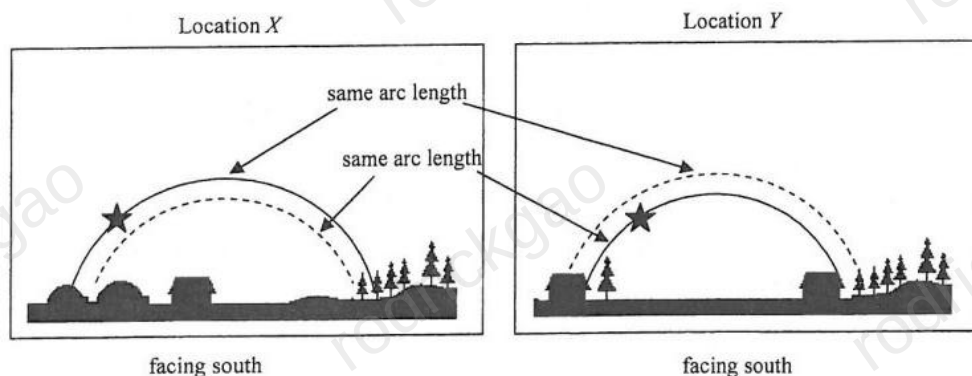
D. 10 pc

Official answer: D

D. Relative to the background, X shifts by two grid squares, or 0.2 arc second. The parallax is half this: $p = 0.1$ arc second. Hence $d = 1/p = 10$ pc.

Question 1.5

The same star is observed at northern-hemisphere locations X and Y on the same night, facing south. The indicated arcs have equal lengths. Which description is correct?



- A. X is south of Y; the star is above the horizon longer at X.
- B. X is south of Y; the star is above the horizon for less time at X.
- C. X is north of Y; the star is above the horizon longer at X.
- D. X is north of Y; the star is above the horizon for less time at X.

Official answer: A

A. The star culminates higher above the southern horizon at X, so X has the lower northern latitude. The visible arc is longer at X; the same star has the same angular rotation rate, so it remains above the horizon longer there.

Question 1.6

Stars P and Q have equal luminosity. P is 25 times as bright as Q. What can be deduced?

- A. P is 5 times as far away as Q.
- B. Q is 5 times as far away as P.
- C. P is 25 times as far away as Q.
- D. Q is 25 times as far away as P.

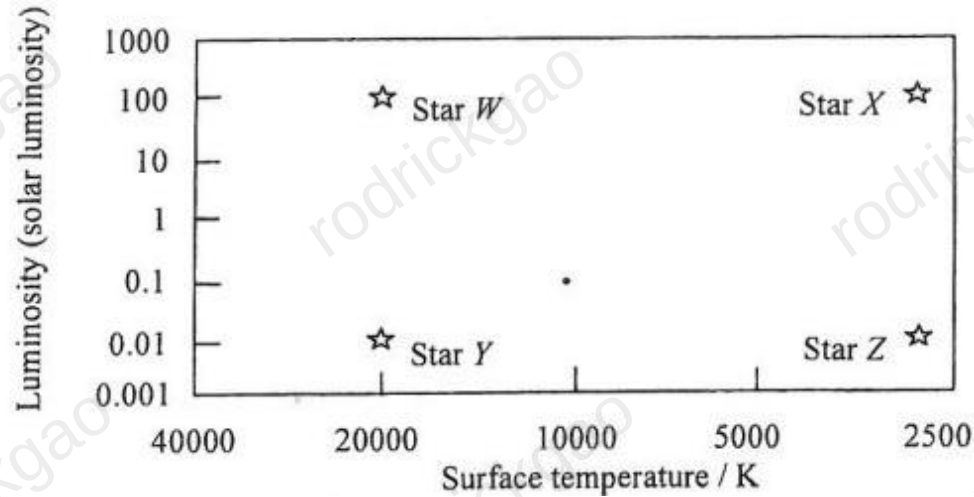
Official answer: B

B. Brightness $b = L/(4\pi d^2)$. Equal L gives $b_P/b_Q = (d_Q/d_P)^2 = 25$, hence $d_Q = 5d_P$.

Question 1.7

The plot gives luminosity and surface temperature for stars W, X, Y and Z. Which statements are correct?

- (1) Radius of X > radius of W.
- (2) Radius of W > radius of Y.
- (3) Radius of Y > radius of Z.



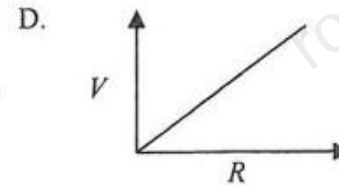
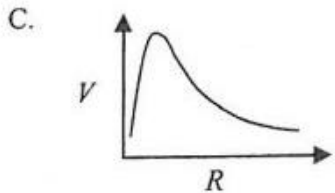
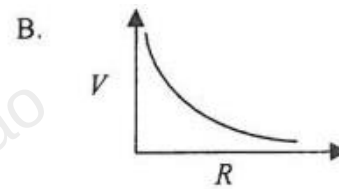
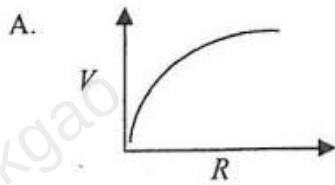
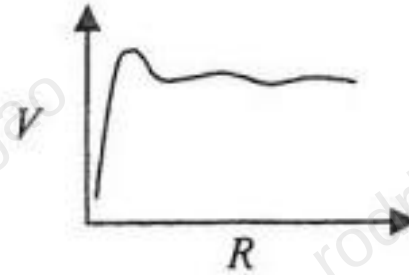
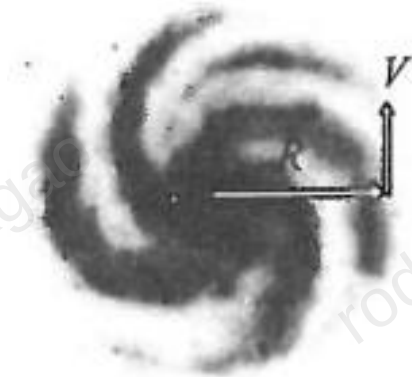
- A. (1) only
- B. (3) only
- C. (1) and (2) only
- D. (2) and (3) only

Official answer: C

C. $L = 4\pi R^2 \sigma T^4$, so $R \propto \sqrt{L/T^4}$. At equal luminosity the cooler star X is larger than W, and Z is larger than Y. At equal temperature W is more luminous and larger than Y. Thus only (1) and (2) hold.

Question 1.8

The galaxy and its observed rotation-speed V versus radius R curve are shown. The curve suggests dark matter. Which rotation curve is expected without dark matter?



- A. Speed rises and levels off.
- B. Speed decreases throughout from a large central value.
- C. Speed first rises, then decreases at large radius.
- D. Speed increases linearly with radius.

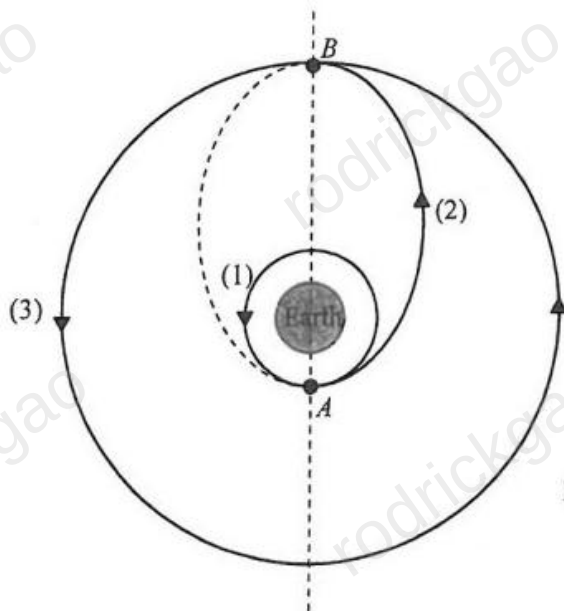
Official answer: C

C. Inside the luminous galaxy, enclosed mass initially increases with radius and speed rises. Outside most luminous matter, M is nearly constant and $V = \sqrt{GM/R}$ falls as $R^{-1/2}$. A flat outer curve instead requires additional unseen mass.

Question 1 Structured Question

Given $GM = 4.0 \times 10^{14} \text{ N m}^2 \text{ kg}^{-1}$, Earth's mean radius = 6400 km and geostationary orbital radius $\approx 42\,400$ km (36 000 km above the surface). G is the gravitational constant and M is Earth's mass.

A rocket first launches a satellite into circular near-Earth orbit (1), 300 km above the surface. At A, a brief engine firing sends it into elliptical transfer orbit (2), whose major axis is AB. At B, another brief firing sends it into geostationary orbit (3).



The three orbits are coplanar; the ellipse touches the two circles at A and B. The engine is off during travel from A to B.

- (a) State and explain the advantage of putting communications satellites in geostationary orbit. (2 marks)
- (b) Find the satellite's speed in near-Earth orbit (1). (2 marks)
- (c)(i) Show that a satellite of mass m in a circular orbit of radius r has total mechanical energy $-GMm/(2r)$. Take gravitational potential energy at infinity as zero. (2 marks)
- (c)(ii) Hence calculate the energy required to transfer a 2000 kg satellite from orbit (1) through A to orbit (3) through B. (2 marks)
- (c)(iii) How long does travel from A to B along transfer orbit (2) take? (2 marks)

Official marking scheme

Marks

1. (a) Satellites will be directly above a certain location on the equator of the Earth, with period = 24 hrs same as that of the Earth, thus enables easy transmitting / receiving signals from the Earth / no altering of aerial for tracking the satellite is required.

1A

1A 2

(b) $\frac{mv^2}{r} = \frac{GMm}{r^2}$

1M

$$v = \sqrt{\frac{GM}{r}} = \sqrt{\frac{4.0 \times 10^{14}}{(6.4 \times 10^6 + 0.3 \times 10^6)}}$$

$$= 7727 \text{ m s}^{-1}$$

1A 2

(c) (i) Total energy = $\frac{1}{2}mv^2 + \left(\frac{-GMm}{r}\right)$

1M

$$= \frac{GMm}{2r} + \left(\frac{-GMm}{r}\right) = \frac{-GMm}{2r} \quad \left(\frac{mv^2}{r} = \frac{GMm}{r^2} \text{ i.e. } \frac{mv^2}{2} = \frac{GMm}{2r}\right)$$

1M 2

(ii)

$$\Delta E = \frac{-GMm}{2} \left(\frac{1}{r_B} - \frac{1}{r_A}\right) = \frac{1}{2}(4.0 \times 10^{14})(2000)\left(\frac{1}{6700} - \frac{1}{42400}\right) \times 10^{-3}$$

$$= 5.03 \times 10^{10} \text{ J}$$

1M

1A 2

(iii) Kepler's third law for elliptical orbit $T^2 = \frac{4\pi^2 a^3}{GM}$

$$a = [r_A + r_B] \div 2$$

$$= \frac{6.7 \times 10^6 + 42.4 \times 10^6}{2} \text{ m}$$

$$= 2.455 \times 10^7 \text{ m}$$

1M

$$\text{Time from A to B} = \frac{T}{2} = \frac{1}{2} \sqrt{\frac{4\pi^2 a^3}{GM}} = \frac{2\pi}{2} \sqrt{\frac{a^3}{GM}} = \pi \sqrt{\frac{(2.455 \times 10^7)^3}{4.0 \times 10^{14}}}$$

$$= 19107 \text{ s} = 318.5 \text{ min} / 5.3 \text{ hrs}$$

1A 2

{Or: $T^2 \propto a^3$

$$\left(\frac{T}{24}\right)^2 = \left[\frac{(6700 + 42400) \div 2}{42400}\right]^3$$

$$T = 10.6 \text{ hrs} \Rightarrow t = 5.3 \text{ hrs}$$

(a) Matching Earth's angular speed and direction in the equatorial plane makes the satellite stationary in the ground frame, so a fixed dish keeps pointing at it.

(b) Use distance from Earth's centre, not altitude: $r_A = (6400 + 300) \text{ km} = 6.7 \times 10^6 \text{ m}$. Gravity supplies centripetal force, giving $v = \sqrt{GM/r_A} = 7.727 \times 10^3 \text{ m s}^{-1}$.

(c)(i) The same force equation gives kinetic energy $GMm/(2r)$; adding the negative potential energy $-GMm/r$ leaves $-GMm/(2r)$.

(c)(ii) The final circular orbit has less-negative total energy. The two engine burns together must supply $E_B - E_A = 5.03 \times 10^{10} \text{ J}$; energy stays constant during the coast.

(c)(iii) $AB = r_A + r_B$ is the ellipse's full major axis. Its semimajor axis is $2.455 \times 10^7 \text{ m}$. A and B are opposite apsides, so the coast is half a Kepler period: $\pi\sqrt{a^3/GM} = 1.9107 \times 10^4 \text{ s} \approx 5.3 \text{ h}$.

Section B Atomic World

Question 2.1

In an α -particle scattering experiment, atomic electrons have negligible effects on the paths of incident α -particles. What is the most probable reason?

- A. Electrons are so small that α -particles do not collide with them.
- B. Electrons are uniformly distributed inside an atom, so the resultant force on an α -particle is zero.
- C. There is no electrical interaction between electrons and α -particles.
- D. The kinetic-energy change of α -particles colliding with electrons is negligible.

Official answer: D

D. An α -particle is thousands of times more massive than an electron. The maximum fractional energy transfer in an elastic collision is $4m_e m_\alpha / (m_e + m_\alpha)^2 \approx 4m_e / m_\alpha$, which is very small. Electrical interaction still exists.

Question 2.2

According to classical electromagnetic theory, what follows from Rutherford's atomic model?

- A. Stable atoms; continuous spectra.
- B. Stable atoms; line spectra.
- C. Unstable atoms; continuous spectra.
- D. Unstable atoms; line spectra.

Official answer: C

C. An orbiting electron accelerates and should radiate continuously. It loses energy and spirals inward, so the atom is unstable; the continuously changing orbital frequency predicts a continuous spectrum.

Question 2.3

Which spectra are continuous?

- (1) A burning candle.
- (2) An incandescent lamp.
- (3) A gas discharge tube.

- A. (1) only
- B. (3) only
- C. (1) and (2) only
- D. (2) and (3) only

Official answer: C

C. Hot soot particles in a candle and the hot solid lamp filament give thermal continua. Excited low-pressure gas emits characteristic discrete lines.

Question 2.4

Hydrogen electron energy levels are $E_n = -E_0/n^2$, where E_0 is constant and $n = 1, 2, 3, \dots$. What is the maximum wavelength of a photon able to ionize hydrogen in its first excited state? h is Planck's constant and c is the speed of light in vacuum.

- A. $3hc/(4E_0)$
- B. hc/E_0
- C. $4hc/(3E_0)$
- D. $4hc/E_0$

Official answer: D

D. The first excited state has $n = 2$ and $E = -E_0/4$. Threshold ionization needs $hc/\lambda = 0 - (-E_0/4)$, giving $\lambda_{\max} = 4hc/E_0$. Both the English and Chinese original questions specify the first excited state.

Question 2.5

When an electron in hydrogen falls from an excited state to the ground state, which radiation will NOT be emitted?

- A. γ -radiation
- B. Ultraviolet radiation
- C. Infrared radiation
- D. Visible light

Official answer: A

A. A return may occur through several intermediate levels, emitting infrared, visible and ultraviolet photons. Hydrogen electronic transitions have energies below 13.6 eV, far below γ -ray energies.

Question 2.6

Figure (1) gives line spectra of elements X, Y and Z. Figure (2) is the spectrum of a mineral. Which element must be absent from the mineral?

Figure (1)

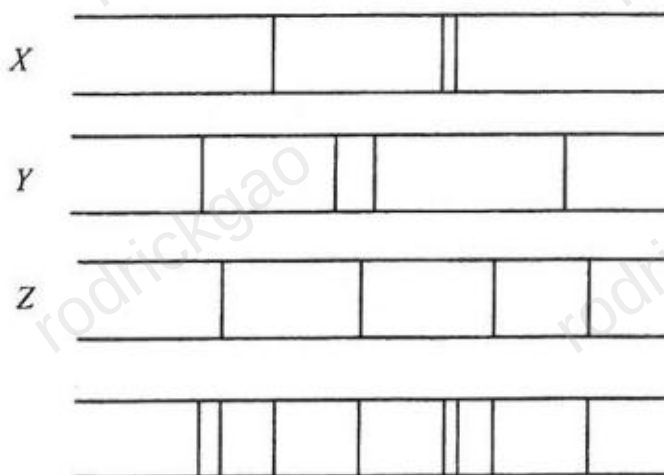


Figure (2)

- A. X
- B. Y
- C. Z
- D. All three elements are present.

Official answer: B

B. Match line positions, not just the number of lines. All X and Z lines occur in the mineral spectrum, whereas characteristic Y lines (including its rightmost line) are missing.

Question 2.7

A typical transmission electron microscope (TEM) resolves about 0.2 nm. Particles with the same charge as an electron but four times its mass are accelerated through the same voltage. What is the new minimum resolvable length?

- A. 0.05 nm
- B. 0.1 nm
- C. 0.4 nm
- D. 0.8 nm

Official answer: B

B. At fixed charge and accelerating voltage, $K = qV$ is unchanged. Momentum $p = \sqrt{2mK}$ doubles when mass quadruples, so $\lambda = h/p$ halves. Resolution length scales with wavelength: $0.2/2 = 0.1$ nm.

Question 2.8

A cube of side 1 mm is divided into cubes of side 1 nm. By what factor does the total surface area increase?

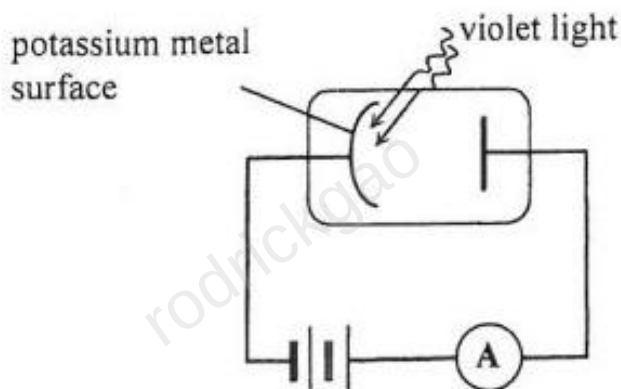
- A. 10^6
- B. 10^8
- C. 10^{10}
- D. 10^{12}

Official answer: A

A. Each edge is divided into $N = 10^{-3}/10^{-9} = 10^6$ pieces. There are N^3 cubes, each with $1/N^2$ of the original surface area, so the total factor is $N = 10^6$.

Question 2 Structured Question

Violet light of a certain wavelength illuminates a potassium surface. The maximum emitted-electron kinetic energy is 0.81 eV; potassium's work function is 2.30 eV.



(a)(i) Find the energy of a violet photon in eV. (1 mark)

(a)(ii) Explain why not all emitted electrons have the maximum kinetic energy. (1 mark)

The violet-light intensity is 0.01 W m^{-2} .

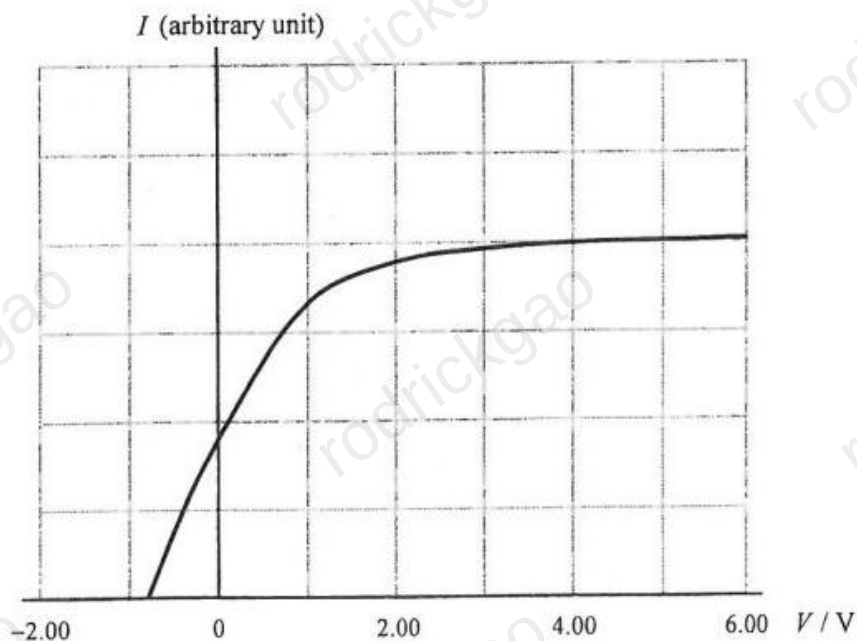
(b)(i) Classical wave theory requires an atom to absorb sufficient wave energy before emitting an electron.

Estimate the minimum absorption time for a potassium atom whose effective absorbing area is 0.01 nm^2 . Take $1 \text{ nm} = 10^{-9} \text{ m}$. (2 marks)

(b)(ii) Explain the almost-zero observed delay in photoemission even with very weak light. (1 mark)

(c) If the illuminated potassium area is $4.00 \times 10^{-4} \text{ m}^2$, how many photons strike it per second? Find the maximum photoelectric current if one electron is emitted per 10 incident photons. (3 marks)

(d) The graph shows photoelectric current I against cathode–anode potential difference V . Copy it and use a dotted line to sketch the curve when light intensity is halved. (2 marks)



Official marking scheme

Marks

2. (a) (i) $E = hf = \text{work function} + KE_{\text{max}}$
 $= 2.30 \text{ eV} + 0.81 \text{ eV} = 3.11 \text{ (eV)}$ 1A 1

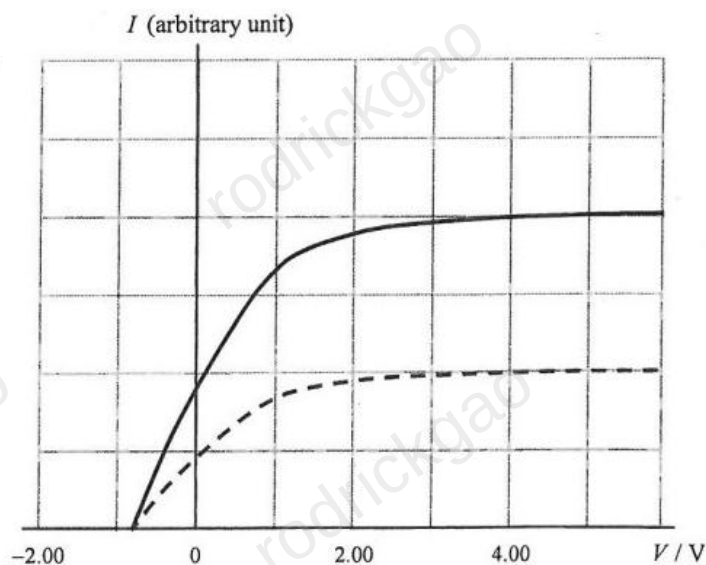
- (ii) Only those conduction / free electrons at the surface can have the maximum kinetic energy. 1A
Or The work function of a metal is only the minimum energy required to eject an electron.
Or The conduction / free electrons in metal have different energies.
Or Less energetic electrons are tightly bound to the nuclei and require more energy to break free of its attraction to the nuclei.
Or Some electrons are not at the surface of metal so don't have maximum kinetic energy. 1

(b) (i) Energy absorbed by an atom = work function
 $(0.01 \text{ W m}^{-2}) \times [0.01 \times (10^{-9})^2 \text{ m}^2] \times t \text{ s} = 2.30 \times (1.60 \times 10^{-19}) \text{ J}$
 $t = 3680 \text{ s} = 61.3 \text{ min.}$ 1M
 1A 2

- (ii) If a single photon has sufficient energy to knock out an electron, the electron gains enough energy in just one collision. 1A
Or It is a one-to-one process / an electron can be ejected instantaneously if it accepts a photon of energy larger than the work function of the metal. 1

(c) $(0.01 \text{ W m}^{-2}) \times (4.00 \times 10^{-4} \text{ m}^2) \div [3.11 \times (1.60 \times 10^{-19}) \text{ J}]$
 $= 8.04 \times 10^{12} \text{ (photons per second)}$ 1A
 $(8.04 \times 10^{12}) \times 0.1 \times (1.60 \times 10^{-19}) \text{ A}$
 $= 1.29 \times 10^{-7} \text{ A} = 0.13 \text{ } \mu\text{A}$ 1M
 1A 3

(d)



2

- (a)(i) Energy conservation for one photon gives $3.11 \text{ eV} = 4.976 \times 10^{-19} \text{ J}$. (a)(ii) Work function is the least escape cost; deeper electrons can lose energy before reaching the surface and others have lower initial energies.
- (b)(i) Convert area carefully: $0.01 \text{ nm}^2 = 10^{-20} \text{ m}^2$. The classical received power is 10^{-22} W , so $t = \phi/P = 3.68 \times 10^{-19}/10^{-22} = 3680 \text{ s}$.
- (b)(ii) The photon model does not require gradual accumulation of wave energy. Even at low intensity, each absorbed photon supplies its entire 3.11 eV in one event.
- (c) Incident power is $4.00 \times 10^{-6} \text{ W}$. Divide by photon energy for $8.04 \times 10^{12} \text{ photons s}^{-1}$; multiply by 0.1 and the electron charge to get $1.29 \times 10^{-7} \text{ A}$ when all emitted electrons are collected.
- (d) Frequency and photon energy remain fixed, so K_{max} and the stopping potential magnitude 0.81 V remain fixed. Half the photon flux gives half the emitted-electron rate: $I_{\text{new}}(V) = I_{\text{old}}(V)/2$.

Section C Energy and Use of Energy

Question 3.1

An LED usually produces monochromatic light because

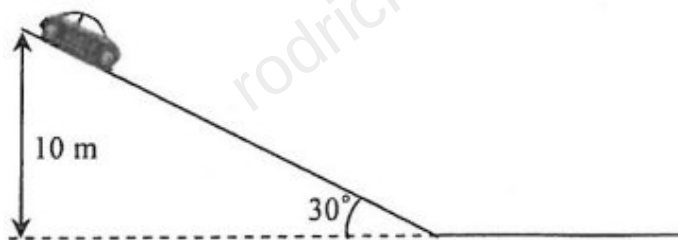
- A. the energy-level difference between the p-layer and n-layer is fixed.
- B. its surface is coated with only one fluorescent material.
- C. its operating temperature is fixed.
- D. it permits current in only one direction.

Official answer: A

A. Electron–hole recombination releases photons of energy approximately equal to the semiconductor band gap. A characteristic gap gives a narrow range of frequencies, $hf \approx E_g$. The option's p-layer/n-layer description is the paper's simplified wording.

Question 3.2

A 1000 kg car has regenerative braking that converts kinetic energy into chemical energy stored in a battery. It descends a 30° slope at constant speed through a vertical height of 10 m. With overall regenerative efficiency 30%, how much energy is stored? $g = 9.81 \text{ m s}^{-2}$.



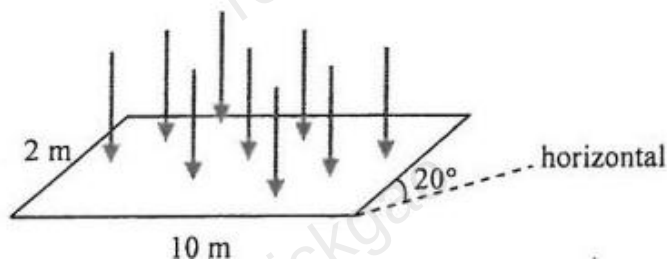
- A. 14.7 kJ
- B. 29.4 kJ
- C. 49.1 kJ
- D. 98.1 kJ

Official answer: B

B. Constant speed means no net kinetic-energy change. The released potential energy is $mgh = 1000 \times 9.81 \times 10 = 98\,100 \text{ J}$. Stored energy $= 0.30mgh = 29\,430 \text{ J} \approx 29.4 \text{ kJ}$.

Question 3.3

Sunlight of intensity 1000 W m^{-2} arrives vertically. What power falls on a solar panel 2 m wide and 10 m long, inclined at 20° to the horizontal?



- A. 6840 W
- B. 7280 W
- C. 18 800 W
- D. 20 000 W

Official answer: C

C. The area normal to the vertical rays is the horizontal projection $A \cos 20^\circ$. Thus $P = 1000 \times (2 \times 10) \cos 20^\circ = 18\,794 \text{ W} \approx 18\,800 \text{ W}$.

Question 3.4

A wind turbine drives a pump lifting water to a reservoir. Blade length is 10 m, average wind speed 5 m s^{-1} , overall efficiency 20%, and air density 1.23 kg m^{-3} . How much water is pumped in 8 h if its mean potential-energy gain is 981 J kg^{-1} ?

- A. 39.4 kg
- B. $2.84 \times 10^4 \text{ kg}$
- C. $1.15 \times 10^5 \text{ kg}$
- D. $1.42 \times 10^5 \text{ kg}$

Official answer: D

D. Swept area $A = \pi(10)^2$. Wind power $= \frac{1}{2}\rho A v^3$. Water mass $= [0.20 \times \frac{1}{2} \times 1.23 \times \pi \times 10^2 \times 5^3 \times (8 \times 3600)]/981 = 1.42 \times 10^5 \text{ kg}$.

Question 3.5

A closed foam box measuring $0.5 \text{ m} \times 0.3 \text{ m} \times 0.4 \text{ m}$, with wall thickness 1 cm, contains melting ice. Estimate the heat-conduction rate into it at room temperature 28°C . Foam thermal conductivity is $0.03 \text{ W m}^{-1} ^\circ\text{C}^{-1}$.

- A. 39.5 W
- B. 79 W
- C. 3950 W
- D. 7900 W

Official answer: B

B. All six faces conduct heat: $A = 2(0.5 \times 0.3 + 0.3 \times 0.4 + 0.5 \times 0.4) = 0.94 \text{ m}^2$. Melting ice is at 0°C . $P = kA\Delta T/d = 0.03 \times 0.94 \times 28/0.01 = 78.96 \text{ W} \approx 79 \text{ W}$.

Question 3.6

Making a building's glass windows smaller can reduce its Overall Thermal Transfer Value (OTTV) because:

- (1) Glass has much higher thermal conductivity than concrete.
- (2) Open windows allow heat transfer by convection.
- (3) Glass permits radiative heat transfer.

- A. (1) only
 B. (2) only
 C. (1) and (3) only
 D. (2) and (3) only

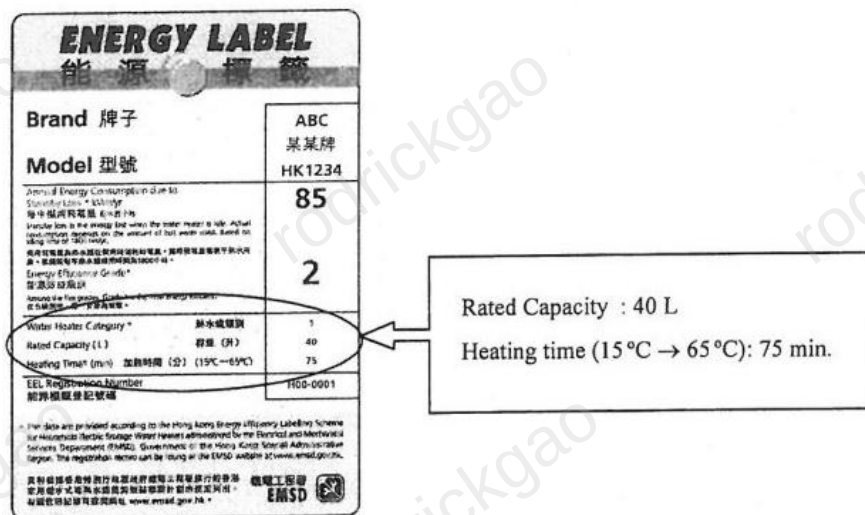
Official answer: D

D. An open window allows warm air to enter and carry thermal energy; glazing also transmits solar radiation. Glass does not have a vastly greater thermal conductivity than concrete, so (1) is not the reason.

Scope qualification: official D uses a broad heat-gain explanation, including convection through open windows and solar radiation through glazing. In the formal OTTV envelope calculation, however, OTTV accounts for conductive and solar heat gains through the building envelope per unit envelope area; ventilation or infiltration heat load from open windows is excluded. Thus (2) explains an additional cooling load, not a component of formal OTTV. The original question and official answer D are preserved.

Question 3.7

The storage-water-heater energy label specifies rated capacity 40 L and heating time 75 min from 15°C to 65°C. Estimate effective output power when filled with 40 L of water initially at 15°C. Use 1 L = 1000 cm³, water density 1000 kg m⁻³ and specific heat capacity 4200 J kg⁻¹ °C⁻¹.



- A. 1870 W
 B. 2430 W
 C. 85 000 W
 D. 112 000 W

Official answer: A

A. Water mass is 40 kg. Useful power = $mc\Delta T/t = 40 \times 4200 \times (65 - 15)/(75 \times 60) = 1867 \text{ W} \approx 1870 \text{ W}$. Use the temperature rise of 50°C and convert minutes to seconds.

Question 3.8

Four hydrogen nuclei, each of mass 1.007825 u, fuse to form a nucleus of element Q of mass 4.002603 u. Which statements are correct?

- (1) The hydrogen nuclei need extremely high kinetic energy to start fusion.
- (2) About 26.7 MeV is released.
- (3) Q is radioactive.

A. (1) only

B. (3) only

C. (1) and (2) only

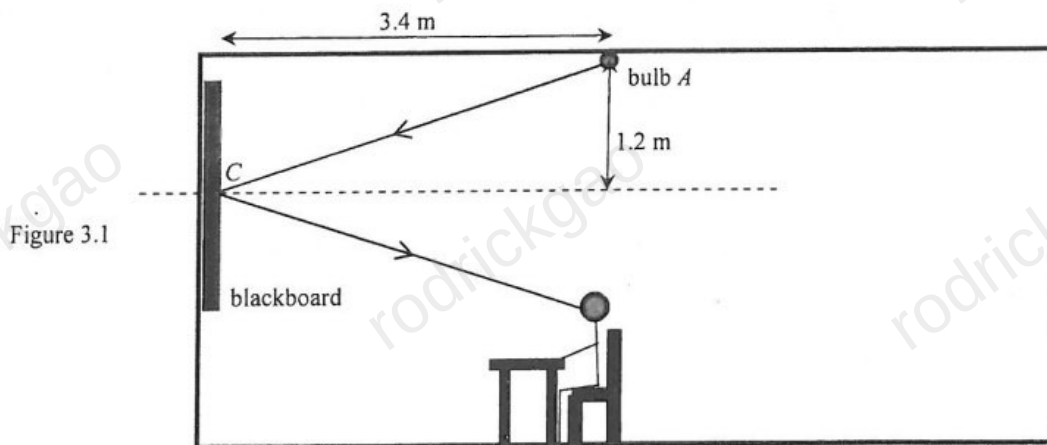
D. (2) and (3) only

Official answer: C

C. High kinetic energy allows nuclei to approach against electrostatic repulsion. With the supplied masses, $\Delta m = 4(1.007825) - 4.002603 = 0.028697$ u; $\Delta E \approx 0.028697 \times 931.5 = 26.7$ MeV. Q is stable helium-4, so (3) is false. The quoted masses are atomic-mass values despite the source calling them nuclear masses; retain the given values for this calculation.

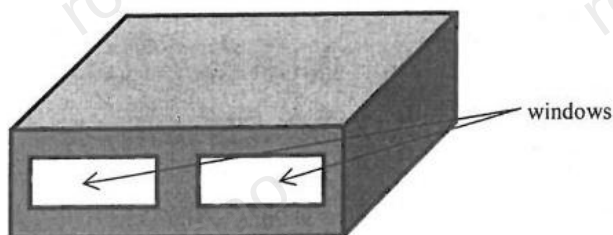
Question 3 Structured Question

Figure 3.1 shows a classroom with an incandescent bulb A of luminous flux 2000 lm. Treat it as a point source. A is 3.4 m horizontally from the blackboard and 1.2 m above its centre C.



- Find the illuminance at C due to A, in lm m^{-2} , neglecting reflected light. (2 marks)
- A mainly illuminates a student's desk, but the ray reflected into the student's eyes is undesirable. Explain which blackboard surface reduces this problem. (2 marks)
- Figure 3.2 shows the classroom exterior; average heat gain from outside is 14.5 kW.

Figure 3.2



The classroom holds at most 15 people, each producing 100 J of heat per second. Its six identical incandescent bulbs each produce 80 J of heat per second.

- Estimate the required air-conditioner cooling capacity in kW, accounting for internal heat and heat from outside. Assume no other heat-producing equipment. (2 marks)
- Each bulb has electrical power rating 100 W. The air conditioner consumes 0.5 J of electricity per 1 J of heat removed. Estimate the monthly electricity cost for lighting and air conditioning, for 8 h/day and 20 days/month, at \$1.0/kW h. (3 marks)
- Suggest one building or appliance change that lowers energy use and electricity cost. (1 mark)

Official marking scheme

Marks

3. (a) $2000 \left[\frac{1}{4\pi(3.4)^2} \cos^3 \left(\tan^{-1} \left(\frac{1.2}{3.4} \right) \right) \right]$
 $= 11.5 \text{ (lm m}^{-2}\text{)}$

1M

1A 2

- (b) Rough surface should be used such that reflection becomes diffuse to reduce glare.

1A

1A 2

- (c) (i) $14.5 \text{ kW} + 15 \times 0.1 \text{ kW} + 6 \times 0.08 \text{ kW}$
 $= 16.48 \text{ (kW)}$ (accept 16.48 kW or 16.5 kW)

1M

1A 2

- (ii) $(6 \times 0.1 \text{ kW} + 16.48 \text{ kW} \times 50\%)$
 $\times 8 \times 20 \times 1.0$
 $= \$ 1414.4$ (accept \$ 1414.4 or \$1416)

1M

1M

1A 3

- (iii) Windows with low-e coating.
Or Thicker walls.
Or Replace light bulb by fluorescent lamp.
Or Replace air-conditioner with higher cooling capacity / COP.

1A

1

(a) Let $\theta = \tan^{-1}(1.2/3.4)$ and $s = \sqrt{(3.4^2 + 1.2^2)}$. Isotropic intensity is $2000/(4\pi)$ cd. The board receives $E = [2000/(4\pi s^2)] \cos \theta = 11.54 \text{ lm m}^{-2}$; the cosine accounts for oblique incidence.

(b) Microscopic surface normals on a rough board point in many directions, spreading reflected light rather than directing a strong specular ray into the eye.

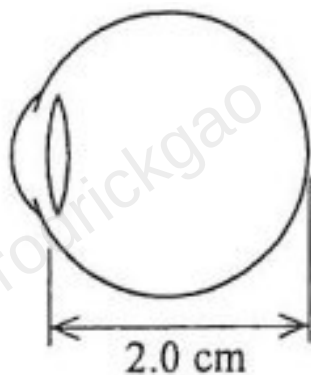
(c)(i) At full occupancy, total heat load is $14.5 + 1.50 + 0.48 = 16.48 \text{ kW}$. (c)(ii) Cooling electrical input is $0.5 \times 16.48 = 8.24 \text{ kW}$; lamps draw 0.60 kW . Over 160 h, energy = $8.84 \times 160 = 1414.4 \text{ kW h}$, costing \$1414.4.

(c)(iii) For example, use a higher-COP air conditioner: for the same heat load its electrical input $\dot{Q}/\text{COP}$ is smaller. Higher cooling capacity alone, although included in the official wording, does not guarantee lower consumption.

Section D Medical Physics

Question 4.1

The eye shown has 2.0 cm between the retina and the optical centre of its refracting parts. Their minimum power is +55 D. What spectacle power corrects this defect?



- A. -5 D
- B. -10 D
- C. +5 D
- D. +10 D

Official answer: A

A. Parallel rays must focus 0.020 m behind the optical centre, requiring total power $1/0.020 = +50$ D. The relaxed eye is 5 D too strong, so a diverging spectacle lens supplies $50 - 55 = -5$ D (thin lenses treated as in contact).

Question 4.2

Jacky observes two point objects separated by 5 mm, emitting green light of wavelength 550 nm. His pupil diameter in normal daylight is about 3 mm. Estimate the greatest distance at which he can resolve the objects.

- A. 42.4 m
- B. 24.2 m
- C. 22.4 m
- D. 20.4 m

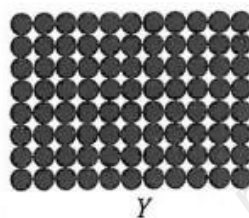
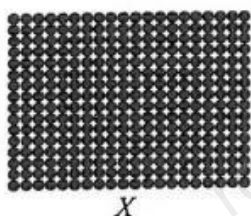
Official answer: C

C. Rayleigh's criterion gives $\theta_{\min} = 1.22\lambda/D$. With angular separation s/L , $L_{\max} = sD/(1.22\lambda) = (5 \times 10^{-3})(3 \times 10^{-3})/(1.22 \times 550 \times 10^{-9}) = 22.4$ m.

Question 4.3

Coherent fibre bundles X and Y are used in endoscopes. Their cross-sections have the same dimensions, but X contains more, finer fibres. Which statements are correct?

- (1) X gives a much brighter image.
- (2) X can be bent more than Y.
- (3) X gives higher image resolution.



- A. (1) and (2) only
- B. (1) and (3) only
- C. (2) and (3) only
- D. (1), (2) and (3)

Official answer: C

C. Finer fibres are more flexible, and more fibres provide more independent picture elements across the same image area. The overall collecting area is comparable, so merely subdividing it does not make the image much brighter.

Question 4.4

The ear is sensitive because sound-pressure changes are greatly amplified before reaching the inner ear. Which facts contribute to this amplification?

- (1) Ear bones act as levers when transmitting vibrations from the eardrum to the oval window.
- (2) The eardrum area is much larger than the oval-window area.
- (3) Inner-ear fluid is much denser than outside air.

- A. (1) and (2) only
- B. (1) and (3) only
- C. (2) and (3) only
- D. (1), (2) and (3)

Official answer: A

A. Ossicle lever action increases force. Applying that force to the much smaller oval window increases pressure further, since $p = F/A$. The fluid's higher density explains the need for impedance matching but does not itself provide these mechanical amplification factors.

Question 4.5

A loudspeaker supplied with 50 W produces sound level 100 dB at a location. There are no other sound sources and electrical-to-sound efficiency is fixed. What input power gives 110 dB at the same location?

- A. 52 W
- B. 55 W
- C. 100 W
- D. 500 W

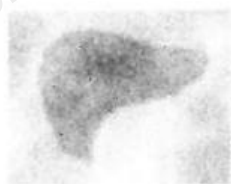
Official answer: D

D. $\Delta\beta = 10 \log_{10}(I_2/I_1) = 10$ dB, so intensity rises tenfold. At fixed efficiency and distance, $I \propto P$; $P_2 = 10 \times 50 = 500$ W.

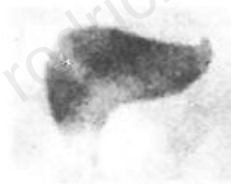
Question 4.6

Technetium-99m undergoes γ -decay with half-life 6 h. It is combined with a substance readily absorbed by the liver and taken by a patient. A gamma camera records images 1, 3 and 6 h later. Which statements are correct?

- (1) Darker regions correspond to greater γ -ray attenuation in the liver.
- (2) The series provides functional information about the liver.
- (3) Differences between images are solely due to radioactive decay.



1 hour after intake



3 hours after intake



6 hours after intake

- A. (1) only
- B. (2) only
- C. (1) and (3) only
- D. (2) and (3) only

Official answer: B

B. This is an emission image: darker regions represent more detected photons from tracer uptake, not greater attenuation of an external beam. The changing tracer distribution reflects liver uptake and processing as well as decay, so it supplies functional information.

Question 4.7

Which statements about ultrasound imaging are correct?

- (1) Ultrasound is potentially hazardous because it is ionizing radiation.
- (2) It is unsuitable for lung scanning because it is almost totally reflected at tissue–air boundaries.
- (3) High-frequency ultrasound penetrates farther but gives lower resolution.

A. (1) only

B. (2) only

C. (1) and (3) only

D. (2) and (3) only

Official answer: B

B. Ultrasound is a mechanical, non-ionizing wave. The large tissue–air acoustic-impedance mismatch causes strong reflection. Higher frequency gives shorter wavelength and better resolution, but greater attenuation and less penetration.

Question 4.8

A car-accident patient is suspected to have internal bleeding in the brain. Which listed imaging method is most suitable for locating the bleeding?

A. Ultrasound scanning

B. Endoscopy

C. X-ray radiography

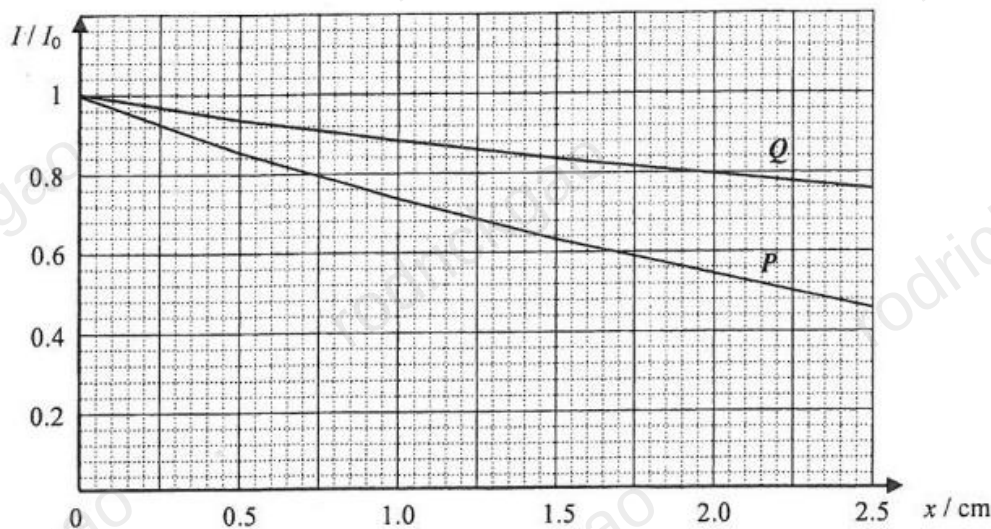
D. Computed tomography (CT)

Official answer: D

D. CT reconstructs cross-sectional attenuation maps and can locate bleeding within the brain without superposition of the entire skull and head. A plain radiograph overlaps structures; the other listed methods do not provide suitable views through the intact adult skull.

Question 4 Structured Question

(a) Figure 4.1 shows how the intensity of an X-ray beam changes after distance x through media P and Q. Initial intensity is I_0 .



- (a)(i) Find the half-value thickness of P. (1 mark)
- (a)(ii) Find P's linear attenuation coefficient. (2 marks)
- (a)(iii) Is Q's density higher than, equal to or lower than P's? (1 mark)
- (b) Figure 4.2 is a chest X-ray radiograph.



- (b)(i) Explain how the image forms through the different effects of soft tissue and bone on passing X-rays. (2 marks)
- (b)(ii) Briefly explain why CT gives more detailed structural information than an X-ray radiograph. (2 marks)
- (b)(iii) Give two reasons, other than CT scanners being more expensive, why conventional X-ray radiography has not been completely replaced by CT. (2 marks)

Official marking scheme

			Marks
4.	(a)	(i) 2.25 cm	1A <u>1</u>
		(ii) $x_{1/2} = \frac{\ln 2}{\mu}$ (or $0.5I_0 = I_0 e^{-\mu x_{1/2}}$)	1M
		$0.0225 = \frac{\ln 2}{\mu}$ $\mu = 30.8 \text{ m}^{-1}$ (accept 30.8 m^{-1} and 31.0 m^{-1})	1A <u>2</u>
		(iii) Medium Q: lower density	1A <u>1</u>
	(b)	(i) Intensity of X-rays is attenuated / absorbed when they pass through a medium. The attenuation / absorption in bone is greater than that in soft tissue. Therefore the film appears lighter under bone / darker under soft tissue.	1A 1A <u>2</u>
		(ii) The X-ray tube and detectors rotate round the patient to take multiple X-ray projections / images. The projections are used to reconstruct / compute / make back projection / combine to form tomographs which contain more information of the body.	1A 1A <u>2</u>
		(iii) - radiation exposure or dosage is much higher for CT scan (8.0 mSv Vs 0.01 mSv for X-ray imaging) - not as mobile or easily accessible as X-ray imaging	1A 1A <u>2</u>
(a)(i) Read x where P crosses $I/I_0 = 0.50$, giving about 2.25 cm. (a)(ii) From $I/I_0 = e^{-\mu x}$, $\mu = -\ln(0.5)/0.0225 = 30.81 \text{ m}^{-1}$. The centimetre reading must be converted to metres.			
(a)(iii) Q transmits more at the same thickness, so it attenuates less. Under the comparison intended by this question, this indicates lower density; attenuation also depends on material composition and X-ray energy.			
(b)(i) More absorption in bone leaves fewer photons reaching the film behind it, so that region is less exposed and lighter. More transmission through soft tissue darkens the corresponding film region.			
(b)(ii) A radiograph collapses all depth information into one projection. Rotating source–detector measurements provide many paths through the body, enabling reconstruction of separate slices and reducing overlap.			
(b)(iii) Multiple projections raise radiation dose, and CT equipment is less portable/accessibly deployed. The numerical doses in the official answer are examples from this examination, not universal doses for every procedure.			